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Classification partie 1

Classification

- Classification is a supervised learning method that involves categorizing objects into classes based on quantitative and/or qualitative features characterizing these objects
- The dataset with known classifications is used to learn the classification rules

Classification

Given a set of training data for examples i=1..n, where each example or individual i is characterized by:

- A feature vector x^i
- The value of its label such that $Y^i \in \{-1,1\}$ (for binary classification)
- $Y^i \in \{1,2,\ldots.k\}$

The goal is :

- Predicting the value y^+ for a new feature vector x^+
- To achieve this, the training data is used to construct a classifier �c in such a way that: $y^+ = c(x^+)$

Classification

Classification can be applied in:

- Medical diagnosis
- Image processing
- Agriculture
- Chemistry
- Geology
- Automatic document processing ...etc

Classification

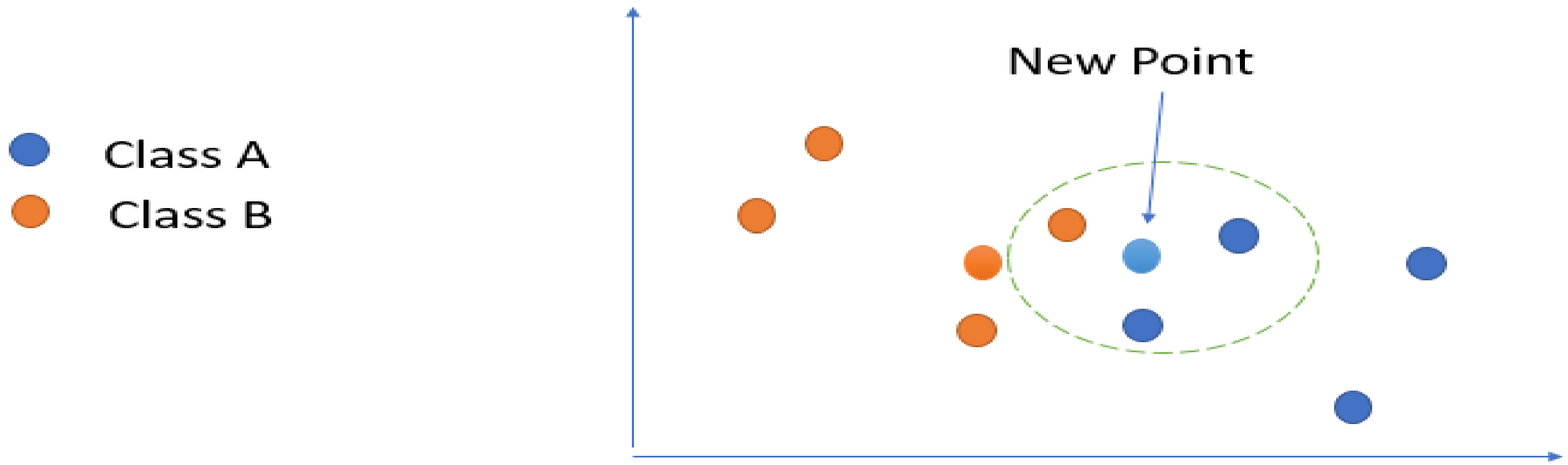
There are several classification algorithms, including:

- K-nearest neighbors
- Support Vector Machines (SVM)
- Decision tree
- Random Forests
- Naive Bayesian model

K-nearest neighbors: KNN

- "K-Nearest Neighbor" (KNN) is one of the simplest supervised machine learning methods
- It allows classifying a new example based on similarity.
- It is a non-parametric algorithm.
- Also known as a "lazy learner"

K-nearest neighbors: KNN



K=3: The three nearest points to the new point are distributed as follows: 2 points belong to class A and 1 point belongs to class B. Therefore, the new point will be classified into class A.

K-nearest neighbors: KNN

There are different methods to calculate the distance between the new point and each training point:

Euclidean distance: the square root of the squared differences between two points.

$$dist(x,y) = \sqrt{\sum_{i=1}^{k} (x_i - y_i)^2}$$

K-nearest neighbors: KNN

- **Manhattan distance:** calculates the difference between two points by using the sum of the absolute values of their differences.

$$dist(x, y) = \sum_{i=1}^{k} |x_i - y_i|$$

- **Hamming distance:** It allows comparing two binary bit strings and represents the number of positions where the bits are different between the two strings.

$$DH(x, y) = \sum_{i=1}^{k} |x_i - y_i|$$

If x= y : DH(x,y)=0

K-nearest neighbors: KNN

The KNN algorithm is described as follows:

Step 1. Select the number k of neighbors.

Step 2. Calculate the distance between the new point and the other points.

Step 3. Take the K nearest neighbors.

Step 4. Among the K neighbors, count the number of elements in each class.

Step 5. Assign the new point to the class for which the number of neighbors is maximal.

K-nearest neighbors: choosing K?

Choosing the value of K in the KNN algorithm is an important decision and can significantly impact the model's performance. Here are some commonly used methods for selecting the optimal value of K:

1.Experimentation:

Try multiple values of K and assess the model's performance on a validation set. Choose the K value that provides the best performance.

2.Grid Search:

Conduct a search over a predefined range of K values and evaluate the model's performance for each value. Choose the K value that offers the best performance.

K-nearest neighbors: choosing K?

3.Square Root of n:

An empirical rule suggests choosing the square root of the total number of observations as the value of K. This can work well in many cases.

4.Elbow Method:

Plot an error curve against different K values. Identify the point where the error ceases to decrease significantly, forming an elbow. This point can be considered a good estimate of K.

K-nearest neighbors: Applicative example

x1	x2	y
7	7	0
7	4	0
3	4	1
1	4	1

3	7	?

- Calculate the distance between example 5 and all the other examples with k=3.

K-nearest neighbors: Applicative example

- Solution:

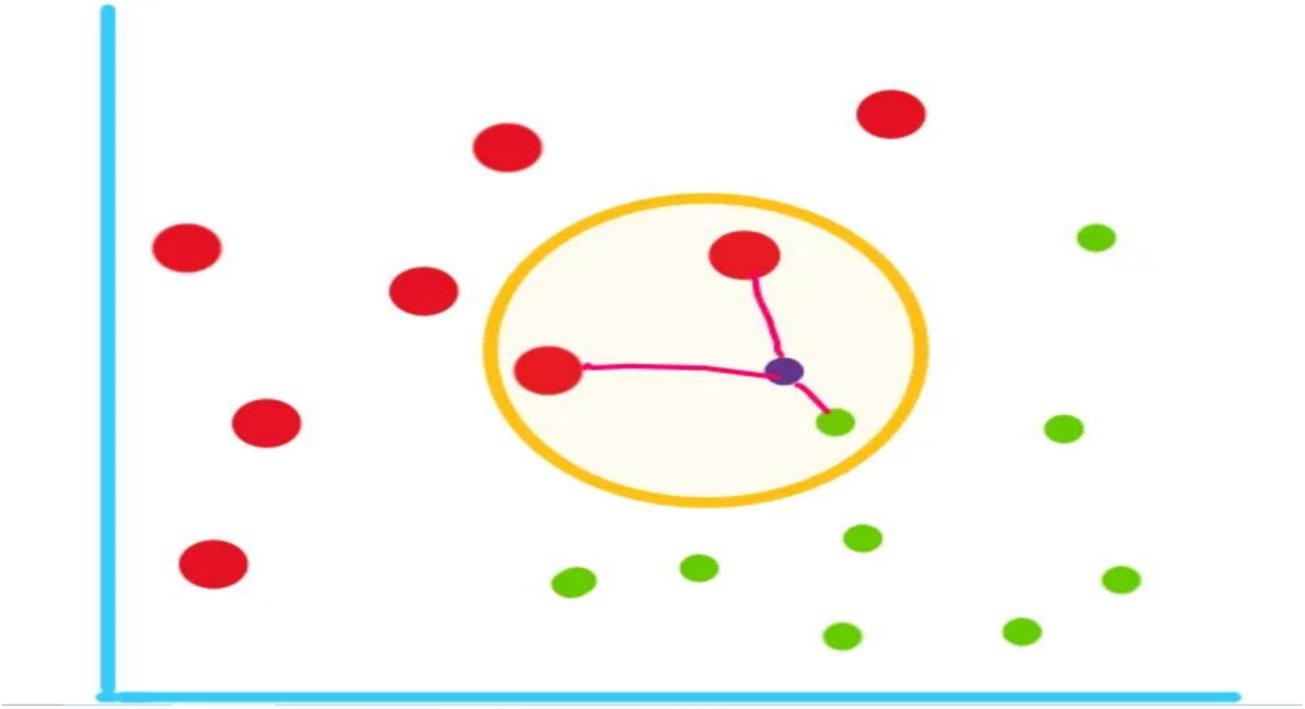
x1	x2	distance
7	7	$(7-3)^2+(7-7)^2=16$
7	4	$(7-3)^2+(4-7)^2=25$
3	4	$(3-3)^2+(4-7)^2=9$
1	4	$(1-3)^2+(4-7)^2=13$

ordre	y
3	0
4	1
1	1
2	1

3	7	1

Weighted K-nearest neighbors:

This basic k-NN algorithm works quite well in many cases, but it treats all neighbors equally.



Weighted K-nearest neighbors:

Steps involved in weighted k-NN:

1. Calculate the distance between the new data point and all data points in the training dataset, just like in basic k-NN.

2. Select the k-nearest data points like in k-NN.

3. Assign a weight to each of the k-nearest neighbors inversely proportional to their distance from the query point. In simpler terms, closer neighbors get higher weights, while farther neighbors get lower weights.

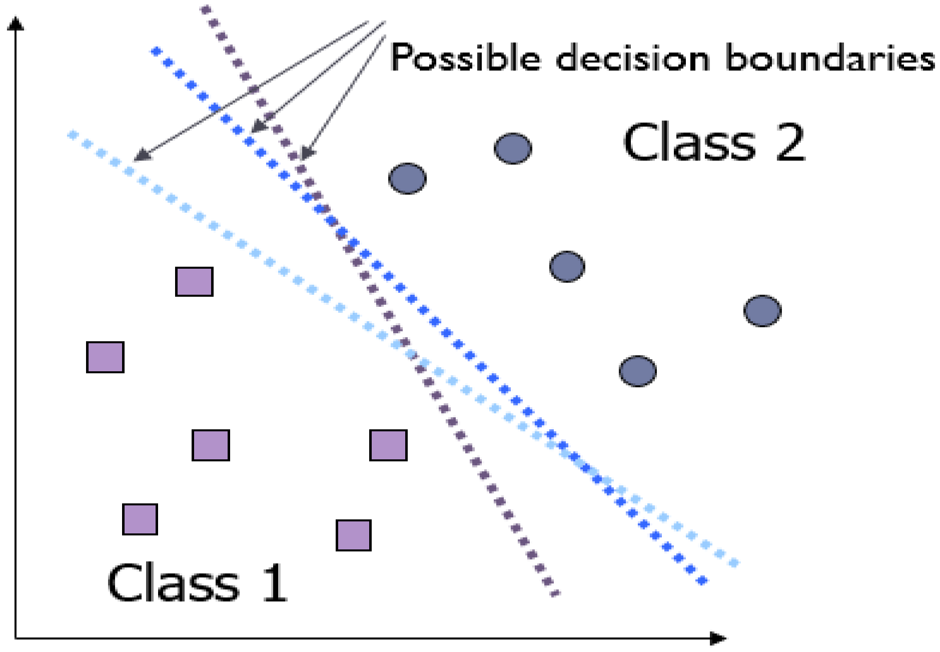
4.For classification tasks, when determining the predicted class, the contributions of neighbors are scaled by their weights. Closer neighbors have a stronger influence on the prediction.

Support Vector Machine: SVM

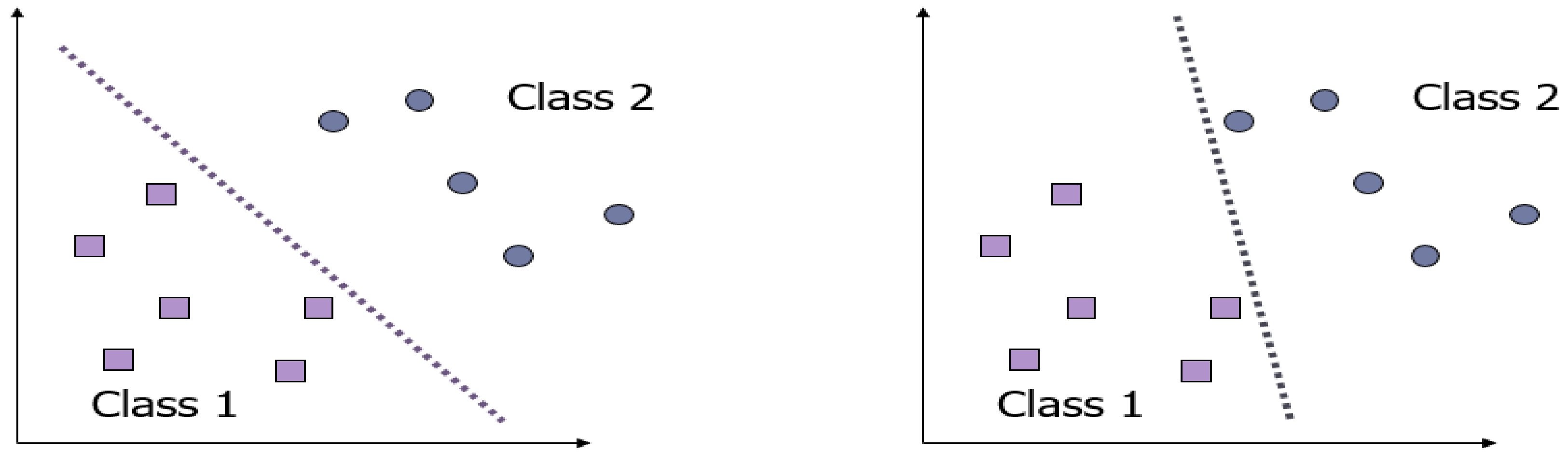
- Support Vector Machine (SVM), also known as separators with large margins, is a supervised method proposed in 1992.
- SVM can be characterized as a supervised learning algorithm capable of solving linear and non-linear binary classification problems.
- It is based on the construction of a hyperplane that separates positive examples from negative examples.

Support Vector Machine

- Multiple decision surfaces exist to separate the classes; which one to choose?



Support Vector Machine

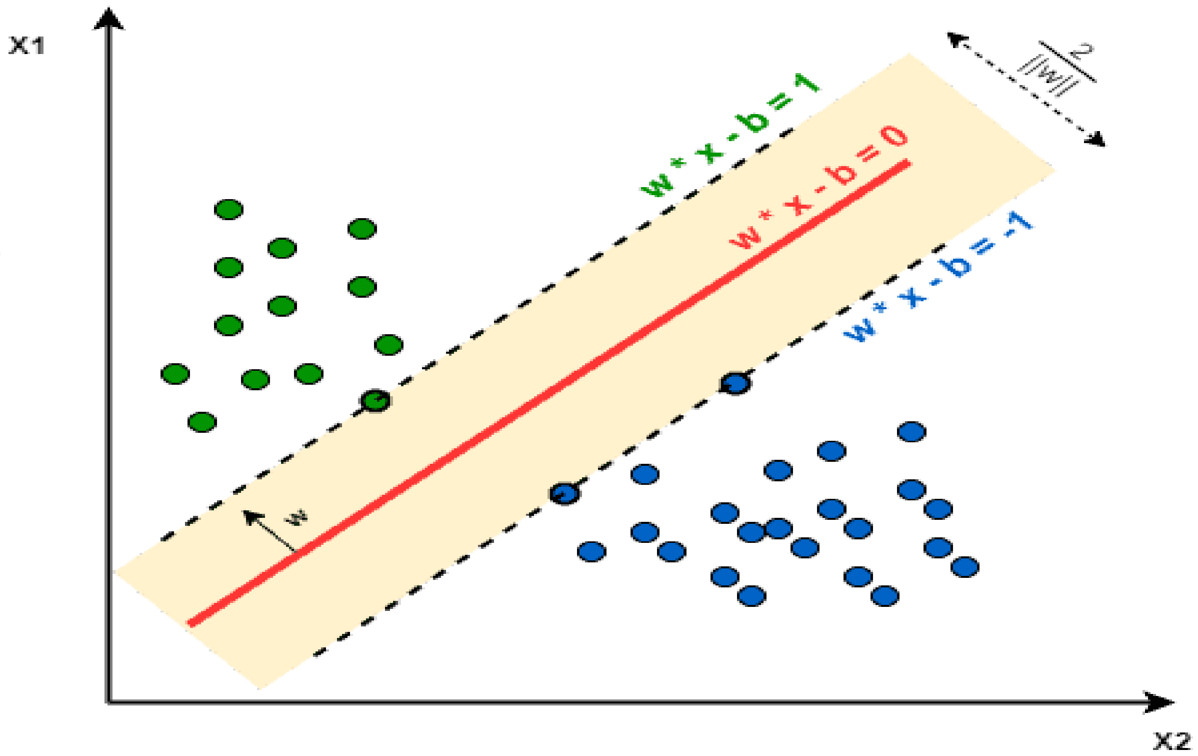


To minimize sensitivity to noise, the decision boundary should be as far away as possible from the data of each class.

Support Vector Machine

- The margin is the minimum distance between the hyperplane and the support vectors
- L'hyperplan est construit à partir des points vecteur de support donnée par l'équation

$$w * x_i + b = 0$$



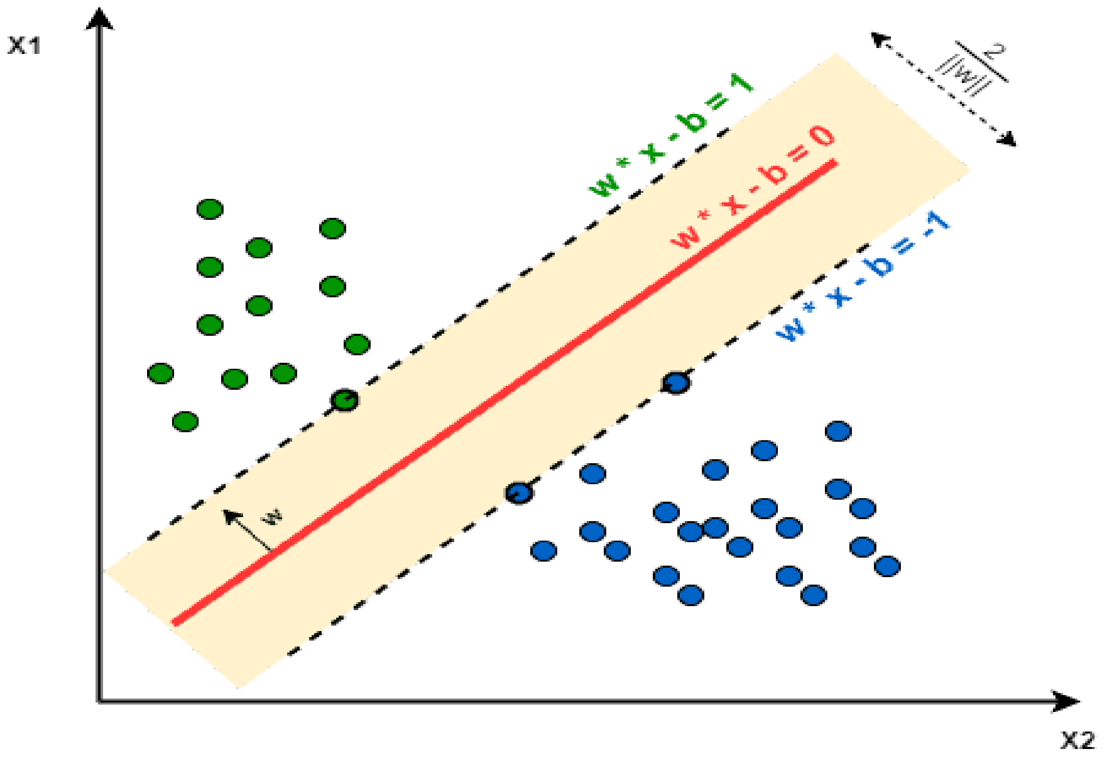
Support Vector Machine

With:

$$w * x_i + b \geq +1 \; si \; y_i = +1$$
$$w * x_i + b \leq +1 \; si \; y_i = -1$$

Les deux contraintes peuvent être combinées comme suit:

$$y_i \; (w * x_i + b) \geq +1$$



Support Vector Machine

- The margin *p* can be calculated as the distance between *H*1 and *H*2.

$$p = \frac{1 - b}{||w||} - \frac{-1 - b}{||w||} = \frac{2}{||w||}$$

- Maximizing the margin *p* is equivalent to minimizing $\frac{||w||}{2}$.
- The optimal hyperplane can be constructed by solving the primal optimization problem: **minimize** $\frac{\mathbf{1}}{\mathbf{2}}\boldsymbol{w}^{\boldsymbol{t}} * \mathbf{w}$ subject to the constraint

$$\forall \boldsymbol{i}\ \boldsymbol{y_i}\,(\boldsymbol{w} * \boldsymbol{x_i} + \boldsymbol{b}) \geq +\mathbf{1}$$

Support Vector Machine

The dual problem is :

$$\boldsymbol{l}(\boldsymbol{w},\boldsymbol{b},\propto) = \sum_{\boldsymbol{i}=\mathbf{1}}^{\boldsymbol{n}} \propto_{\boldsymbol{i}} - \frac{\mathbf{1}}{\mathbf{2}} \sum_{\boldsymbol{i}=\mathbf{1}}^{\boldsymbol{n}} \sum_{\boldsymbol{j}=\mathbf{1}}^{\boldsymbol{n}} \propto_{\boldsymbol{i}} \propto_{\boldsymbol{j}} \boldsymbol{x}_{\boldsymbol{i}} \boldsymbol{x}_{\boldsymbol{j}} \boldsymbol{y}_{\boldsymbol{i}} \boldsymbol{y}_{\boldsymbol{j}}$$

Resolving for *αi* provides the values for the vector **w**.

The class of a new object is obtained by the function:

$$f(x) = signe(\sum_{\boldsymbol{i}=\mathbf{1}}^{\boldsymbol{n}} \propto_{\boldsymbol{i}} \boldsymbol{x}_{\boldsymbol{i}} \boldsymbol{y}_{\boldsymbol{i}} \boldsymbol{x} + \boldsymbol{b})$$

Support Vector Machine

- SVM can be used to classify non-linearly separable data by projecting the data into a higher-dimensional space where linear separation between classes can be achieved using a kernel function.
- The function f is given by: $f(x) = \propto_i K(x_i, x) + C$
- The αi are the weights to be optimized to maximize the separation of class +1 and class -1 as much as possible
- The parameter C is optimized to adjust the training.

Support Vector Machine

There are several kernel functions:

- **Linear kernel:** This is the simplest kernel, given by the inner product ⟨*x*,*y*⟩ plus an optional constant *C*. The linear kernel function is as follows:

$$k(x, y) = (x^t \mathrm{y} + \mathrm{c})$$

Polynomial kernel: This is a non-linear kernel. Polynomial kernels are well-suited for problems where all data is normalized. This type of kernel has two parameters: the degree of the polynomial and the slope alpha. The notation for the function is given by:

$$k(x, y) =\propto (x^t \mathrm{y} + \mathrm{c})^{\,d}$$

Support Vector Machine

- **Gaussian kernel:** It is an example of the radial basis function:

$$k(x, y) = exp(-\gamma||x - y||^2)$$

- **Sigmoid kernel:** Also known as the hyperbolic tangent and multi-layer perceptron (MLP) kernel. It originates from the field of neural networks, where the bipolar sigmoid function is often used as the activation function for artificial neurons.

$$k(x, y) = tanh(\propto x^t \mathrm{y} + \mathrm{c})$$